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Substitution of sugar-sweetened beverages with non-caloric alternatives and weight change: A systematic review of randomized trials and meta-analysis

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Summary

Background: Intake of sugar-sweetened beverages has been associated with weight gain. It is uncertain if replacing an existing use of sugar-sweetened beverages with non-caloric beverages results in long-term reduction in body weight.

Objective: The objective of this study is to explore if a long-term reduction in body weight can be achieved by replacing an existing intake of sugar-sweetened beverages with non-caloric beverages.

Methods: Systematic review and meta-analysis of randomized clinical trials in accordance with PRISMA guidelines. PubMed and EMBASE were searched for literature. Studies with a “substitution” design were included, that is, studies where subjects substituted an existing intake of sugar-sweetened beverages with either artificially sweetened beverages or unsweetened beverages/water. Studies with 6 months or longer follow-up of weight change were included.

Results: Six trials with a total of 1729 participants were included in the meta-analysis. Replacing an existing intake of sugar-sweetened beverages with a non-caloric beverage resulted in a long-term BMI reduction of 0.31 kg/m² compared with the sugar-sweetened beverage-group (95% CI: 0.17–0.44). One study with 1 year's intervention and 2 years follow-up showed a regression towards baseline BMI after the intervention had ended.

Conclusion: Replacing an existing use of sugar-sweetened beverages with artificially sweetened beverages or unsweetened beverages resulted in a long-term 0.31 kg/m² reduction in BMI equivalent to 0.5–1 kg in children and adults, respectively, as long as the interventions lasted.

KEYWORDS

BMI, sugar-sweetened beverages, systematic review, weight change

1 | INTRODUCTION

Intake of sugar-sweetened beverages (SSBs) has been related to high body weight and weight gain.^{1–5} The explanation for this relation may

well be causal as SSBs are easily ingestible calories that may not contribute as much to satiety as solid food does.^{6–10} Therefore, a substitution of SSB with artificially sweetened beverages (ASBs) or water is a potential target for reducing body weight and could be an aim for

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structural prevention of weight gain. However, the relation between energy intake and weight change is complex as it is influenced by both physiological, genetic, socioeconomic, political, behavioral, and environmental factors.^{8,11} Giving public health advice regarding SSB-intake and weight requires robust evidence, preferably from well-designed randomized controlled trials (RCTs) with long follow-up time, as weight loss due to behavioral interventions are most often not maintained in the long-term.¹² Previous systematic reviews have found an association between SSB consumption and weight gain, with some focusing on observational studies,^{13,14} some on RCTs^{3,5} and some on both.^{1,4,15–17} However, many of the RCTs included in the earlier published reviews had short follow-up time (2–12 weeks) and were characterized by being “overfeeding” studies, in which participants who did not habitually consume SSBs were assigned to drinking these (sometimes in relatively large quantities of around 1 L/day) or a control beverage.^{18–23} In this article, we use the term “overfeeding” to describe a study design in which the calorie intake of the intervention group was increased compared with their usual intake.

The present review included studies with a “substitution” design in which participants' habitual SSB-intake was replaced with ASBs or water.^{24–29} The “substitution” design is closer to a real-life setting in which a person with an existing intake of SSB can be advised to replace SSBs with ASBs. Conclusions from meta-analyses are more valid if the included interventions are homogeneous, but the earlier systematic reviews in the field did not distinguish between overfeeding and substitution designs. Furthermore, new RCTs have been published since the latest systematic review.^{27,28} Numerous interventions have shown an effect on body weight in the short term, whereas only few have resulted in weight reduction that was sustained for a longer time.³⁰ From a health perspective short-term effects are not relevant, and the present study included only RCTs with 6 month or longer follow-up time.

Thus, it is uncertain if replacing an existing use of SSBs with non-caloric beverages results in long-term reduction in body weight. The present study set out to answer the question of whether long-term replacement of SSBs with non-caloric beverages result in a sustained weight loss in children and adults.

2 | METHODS

The present systematic literature review and meta-analysis was conducted in accordance with PRISMA guidelines.

2.1 | Literature search

Relevant articles were identified by searching the PubMed (<http://www.ncbi.nlm.nih.gov/pubmed>) and EMBASE (<http://www.embase.com>) from inception (1966 and 1947, respectively) through 5th of April 2022 for RCTs comparing the consumption of SSBs, ASBs, and water with body weight. The search strategy combined terms for sweetening (e.g., sucrose, “sweetening agent”, calori*, sugar*), beverages (e.g., beverage*, soda*, SSB), body weight (e.g., body weight*,

BMI, overweight*, obese*), and intervention studies (e.g., RCT, trial*, intervention*) using both medical subject headings/Emtree and an all-fields search for keywords. Full details on the search strategy are available in Table S1. Reference lists of the included studies and previous systematic reviews were hand searched for additional studies.

2.2 | Study selection

Inclusion of studies for review was based on the following criteria: (1) RCT in humans, (2) the duration of the intervention was 6 months or longer, (3) the RCT addressed consumption of any kind of sweetened beverage normally associated with the term SSB (e.g., sodas, lemonades, energy drinks) but not dairy products such as milk or yogurt, (4) the RCT targeted SSB-intake as the only major difference between study groups, meaning that studies combining reduced SSB-intake and other lifestyle interventions (e.g., low-fat diet or increased exercise) in one study group were not included, (5) the RCT included a measure for body weight (e.g., weight, BMI, BMI z score) in conjunction with the participants' allocation to either intervention- or control group, (6) the study was original research, that is, not a review or a commentary, (7) the participants had an existing intake of SSB and were randomized to replace SSBs with non-caloric beverages or continue drinking SSB.

Titles and abstracts of the identified studies were screened by one author (PT), and potentially relevant studies were selected for full-text review. Cases of doubt were evaluated with the second author (RKR). Covidence software³¹ was used for the screening process. Duplicates were identified and removed using Endnote.³²

2.3 | Data extraction

The following data was extracted from the included articles: general characteristics (author, year, geographic location, sample size, duration), participant characteristics (sex, age, baseline body weight, and BMI), method of assessing SSB intake (e.g., food frequency questionnaire or 24-hour recall), amount of SSB consumed at baseline and follow-up, intervention setting (e.g., home/school delivery of beverages, nutrition counseling), means and standard deviations (SDs) for weight change (body weight, BMI or BMI z score), methods for handling missing data, and funding. The data was manually extracted, tabulated, and summarized in Table 1.

The primary outcome was change in body weight. All available metrics (e.g., kg, BMI units, BMI z score) were extracted. Weight data from baseline and end of intervention were used to compute weight change for the main statistical analysis.

2.4 | Data synthesis and analysis

The primary effect measure was the mean difference in BMI from baseline to the end of the intervention period between intervention and control groups. When data was unavailable, authors were

TABLE 1 Characteristics of studies identified in the systematic literature review. All six studies were included in the meta-analyses.

Study	Country	Participants	Baseline mean age (years)	Baseline mean BMI (kg/m ²)	Duration
de Ruyter 2012 ²⁴	Netherlands	641 Boys = 343 Girls = 298	Intervention: 8.2 Control: 8.2	Intervention: 16.9 Control: 16.8	18 months
Ebbeling 2006 ²⁶	USA (Boston)	103 Boys = 47 Girls = 56	Intervention: 16.0 Control: 15.8	Intervention: 25.7 Control: 24.9	25 weeks
Ebbeling 2012 ²⁵	USA (Boston)	224 Boys = 124 Girls = 100	Intervention: 15.3 Control: 15.2	Intervention: 30.4 Control: 30.1	2 years: 1 year of intervention and 1 year of follow-up
Ebbeling 2020 ²⁷	USA (Boston)	203 Male = 121 Female = 82	SSB (control): 25.9 ASB: 26.7 USB: 27.9	SSB (control): 25.8 ASB: 26.1 USB: 26.6	1 year
Hernández-Cordero 2014 ²⁸	Mexico	240 Male = 0 Female = 240	Intervention: 33.5 Control: 33.3	Intervention: 31.0 Control: 31.0	9 months
Tate 2012 ²⁹	USA (North Carolina)	318 Male = 51 Female = 267	SSB (control): 41.56 ASB: 41.2 USB: 43.2	SSB (control): 36.8 ASB: 36.1 USB: 35.8	6 months

Abbreviations: ASB, artificially sweetened beverage; BMI, body mass index; SSB, sugar sweetened beverage; USB, unsweetened beverage.

TABLE 1 (Continued)

Study	Design	Study question	Intervention	Control
de Ruyter 2012 ²⁴	Parallel, double-blind, randomized intervention	Replacing SSBs with ASBs	One 250 mL can of ASB pr. day delivered weekly at school	One 250 mL can of SSB pr. day delivered weekly at school
Ebbeling 2006 ²⁶	Parallel, randomized intervention	Replacing SSBs with non-caloric beverages (ASBs/water)	Four 350 mL servings pr. day of non-caloric beverages provided through home delivery. Monthly telephone calls and mailing of “Think Before You Drink” refrigerator magnets	Continue usual beverage consumption (which was 380 kcal/day at baseline)
Ebbeling 2012 ²⁵	Parallel, randomized intervention	Replacing SSBs with non-caloric beverages (ASBs/water)	Home delivery of non-caloric beverages every 2 weeks, monthly motivational telephone calls with parents (30 minutes per call), and three check-in visits with participants	Continue usual beverage consumption (which was 324 kcal/day at baseline)
Ebbeling 2020 ²⁷	Parallel, single-blind, randomized intervention	Replacing SSBs with either ASBs or water	Home deliveries of either water or ASBs, specific messages pertaining to beverage consumption, and bi-weekly check-in telephone calls	Home deliveries of SSBs, specific messages pertaining to beverage consumption, and bi-weekly check-in telephone calls
Hernández-Cordero 2014 ²⁸	Parallel, randomized intervention	Replacing SSBs with water	Home delivery or self pick-up of bottled water every 2 weeks. 2–3 L pr. day was provided. They also received nutrition counseling, in which general healthy eating tips were given as well as discouragement of SSBs	Received nutrition counseling, which only included general healthy eating tips with no information about beverages
Tate 2012 ²⁹	Parallel, single-blind, randomized intervention	Replacing SSBs with either ASBs or water	Monthly group meetings in which SSBs were discouraged and where participants received either ASBs or water	Monthly group meetings focused on “Healthy choices” but not beverage consumption

Abbreviations: ASB, artificially sweetened beverage; BMI, body mass index; SSB, sugar sweetened beverage; USB, unsweetened beverage.

contacted and asked to provide data in a suitable format. In cases where authors could not be reached, data was manually transformed to mean BMI change $\pm$ SDs. The process of data transformation is described for the relevant studies in Table S2.

Cochrane's RevMan 5 software³³ was used for the meta-analysis, creating inverse-variance-weighted forest plots to visualize the estimates. The I^2 -statistic was used to evaluate between-study heterogeneity.³⁴ Five prespecified subgroup analyses included analyses of studies that differed in (1) duration of the intervention, (2) participant age (adults/children), (3) baseline BMI, (4) studies with follow-up of weight change after the intervention had ended, and (5) Handling of missing data (imputation versus completers' analysis).

Sensitivity analyses were made by testing the impact of individual studies on the results, creating estimates without each study, one by one. The influence of the method of analysis was investigated by using fixed-effects models for the analysis and comparing these with random-effects models.

2.5 | Risk of bias assessment

The risk of bias of each study was assessed by one author (PT) using Cochrane Collaboration's Risk of Bias Tool 2 (RoB).³⁵ Cases of doubt were evaluated with the second author (RKR). When some characteristics of a study (e.g., details of the intervention) were not described in the main study, the information was usually available in the study

protocol. Publication bias was assessed by creating funnel plots with Cochrane's RevMan 5 software. The risk of publication and other biases were evaluated with the recommendations of Sterne et al.³⁶ in mind: since the meta-analysis included fewer than 10 studies, no tests for funnel plot asymmetry were conducted, rather the funnel plot was visually inspected.

2.6 | Certainty assessment

Certainty in the body of evidence was assessed using the GRADE method (risk of bias, inconsistency, indirectness, imprecision, and other considerations incl. publication bias).

3 | RESULTS

3.1 | Literature search

The search yielded 3409 unique citations, of which 86 were selected for full-text review. Six articles met the inclusion criteria and were included. Figure 1 illustrates the review process. The 2006 study by Ebbeling²⁶ was included despite being 25 weeks long, as it was highly relevant and only 1 week short of duration from meeting the 6 month criterion. Explanations for excluding studies and a review of the findings are described in the [Supporting Information](#).

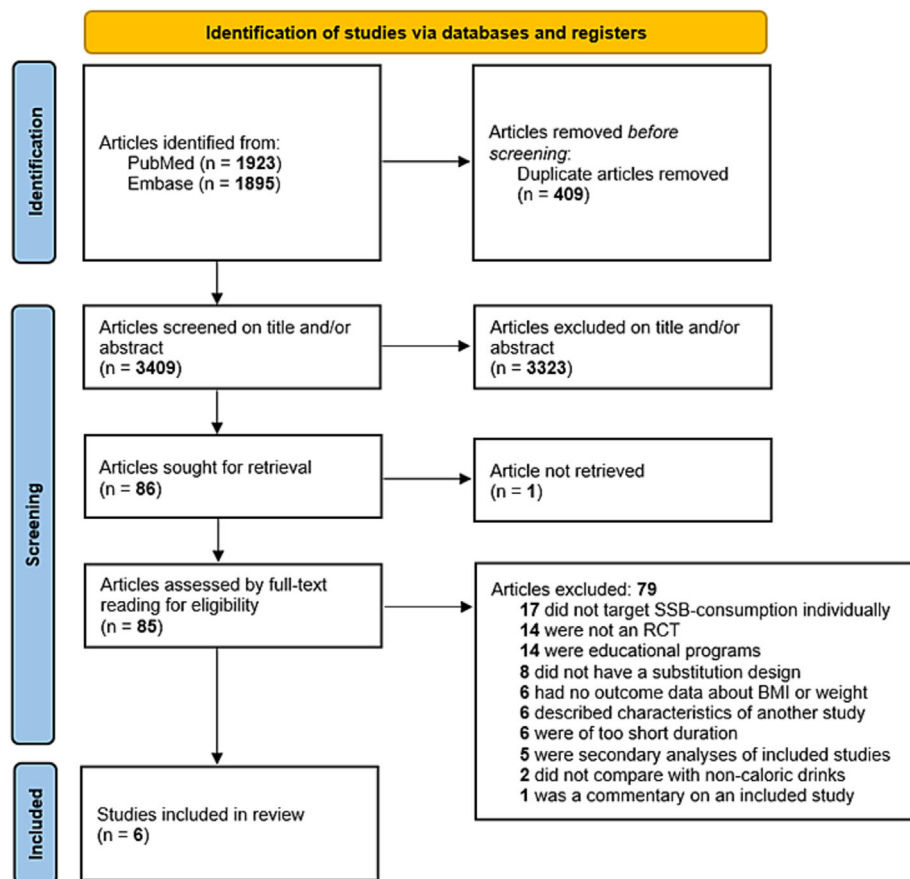


FIGURE 1 Flowchart of systematic literature review and selection process.

3.2 | Study characteristics

Characteristics of the included studies are shown in Table 1. Among the six included studies, four were from the US,^{25–27,29} one was from Mexico²⁸ and one was from the Netherlands.²⁴ The included studies had a “substitution” design and compared SSB consumption to ASB and/or water in a parallel-arm design. Beverages were delivered at home, brought home from school or from monthly meetings with the research staff. Some studies also included nutritional counseling and motivational interventions for all participants such as check-in visits, phone calls and fridge magnets, to increase adherence to the intervention and reduce attrition rates.

3.3 | Bias assessment

The overall risk assessment is shown in Figure 2. The evaluation of each study can be found in Table S3. The studies were overall rated as having a low risk of bias, with half of the studies having some concerns of bias in the “blinding of participants/personnel”-domain, as most studies included a study group drinking water, which was not feasible to blind. In all studies but one,²⁶ some participants were lost to follow-up. In the studies by de Ruyter et al.,²⁴ Tate et al.,²⁹ and Hernández-Cordero et al.,²⁸ missing data were handled with multiple imputation. In two studies by Ebbeling et al., the main result was a completers' analysis. The funnel plot was visually inspected and showed no asymmetry (Figure S2). No substantial bias was identified

regarding industry/funding bias, selective outcome reporting, and selective analysis reporting.

3.4 | SSBs and BMI

Altogether, 1729 participants from the six studies (Table 1) were included in the meta-analysis. Four studies had two arms and compared continuous SSB use with ASB and/or water, and two studies (Ebbeling 2020²⁷ and Tate 2012²⁹) had three arms: SSBs, ASBs, and USBs/water. Ebbeling 2020 found the largest BMI reduction in the water-group, while Tate 2012 found the largest BMI reduction in the ASB-group. The remaining four studies included only one intervention group drinking ASBs and/or water. Therefore, forest plots based on the beverage type were not feasible. Thus, we made two separate meta-analyses and forest plots: one for the largest effect (in which the water group in Ebbeling 2020 and the ASB-group in Tate 2012 was included) and one for the smallest effect (in which the ASB-group in Ebbeling 2020 and the water group in Tate 2012 was included). Replacing SSBs with a non-caloric beverage (ASB and/or water) resulted in mean BMI reductions of 0.28 (95% confidence interval 0.14, 0.42) and 0.31 kg/m² (0.17, 0.44) for the smallest and the largest effect, respectively. Figure 3 displays the forest plot for the largest effect. The forest plot for smallest effect is shown in the Figure S1. Also in the Supporting Information, we display the calculations of how the BMI reduction is equivalent to a weight loss of approximately 0.5 kg in children and 1.0 kg in adults. The

FIGURE 2 Risk of bias-assessment made with Cochrane's revised risk of bias tool.

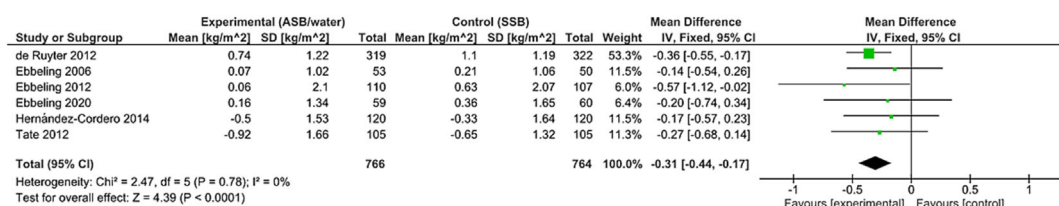
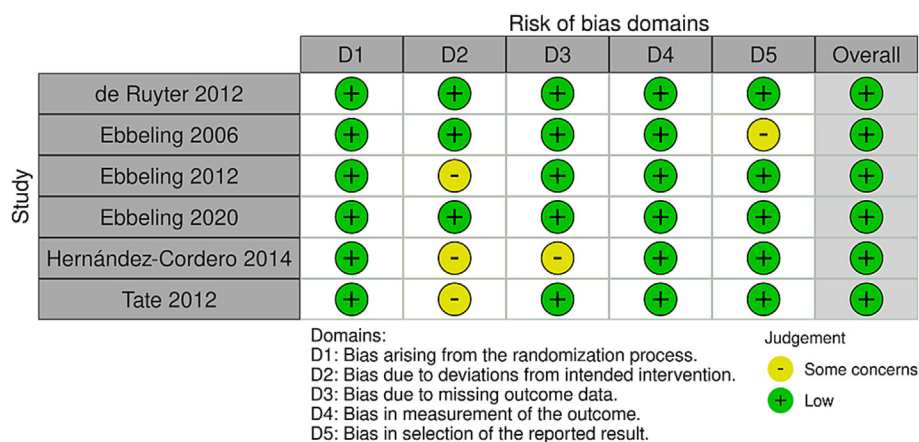


FIGURE 3 Forest plot of BMI (kg/m²) change in control- (SSB-) and intervention- (ASB and/or water-) groups. In studies with multiple intervention groups (either water or ASB), the group that obtained the largest effect is displayed.

between-study heterogeneity as assessed with I^2 was 0% for all analyses. Using a random-effects model did not change the estimated effect size.

3.5 | Prespecified subgroup analyses

The five prespecified subgroup analyses are described below, and the accompanying forest plots can be found in Figures 4 and 5 and in the Figure S1.1–1.10. (1) Duration of the intervention. Arranging the studies from the shortest (25 weeks) to the longest duration (18 months) revealed a trend towards that the three longer interventions had larger effects on BMI than the three shorter interventions: 0.36 kg/m² (0.2, 0.53) and 0.19 kg/m² (–0.04, 0.43), respectively. (2) Participants age. The study by de Ruyter et al.³⁷ included children with a mean age of 8.2 years, while two studies by Ebbeling et al.^{25,26} included teenagers (mean age of 15.2 and 15.9, respectively). A subgroup analysis of these three studies resulted in BMI reduction of 0.34 kg/m² (95% CI: 0.18, 0.50), whereas meta-analysis of the three studies with adult participants resulted in a BMI reduction of 0.22 kg/m² (95% CI: 0.04, 0.47). Figures 4 and 5 display these results. (3) Baseline BMI. Three studies^{25,28,29} were designed to include only participants with a baseline BMI > 30, while three^{24,26,27} included participants regardless of their baseline BMI. Subgroup analyses resulted in BMI reductions of 0.29 kg/m² (0.04, 0.55) and 0.31 kg/m² (0.15, 0.47), respectively. (4) Only one study had follow-up of weight change after the intervention had ended.²⁵ The study included children and the intervention lasted 1 year. After 1 year the body weight was 1.9 kg (SD 0.9 and $p = 0.04$) lower in the intervention group compared with the control group. After 2 years—1 year after the intervention had ended—the difference had diminished to a non-significant

0.8 kg (SD 1.4 and $p = 0.55$). (5) Handling of missing data. In three studies,^{24,28,29} missing outcome data was handled with multiple imputation, while the three studies by Ebbeling et al.^{25–27} conducted a completers' analysis. Studies using multiple imputation found an effect of 0.32 kg/m² (0.16, 0.47), while studies using completers' analysis found an effect of 0.26 kg/m² (–0.01, 0.54).

3.6 | Sensitivity analyses

Sensitivity analyses in which each study was excluded, in turn one by one, generally showed robustness of the results, however, excluding the large study by de Ruyter et al.,²⁴ diminished the BMI difference (largest effect) between study groups to 0.24 kg/m² (95% CI: 0.04, 0.44).

3.7 | Assessments of changes in energy intake

All studies assessed the intake of SSBs at baseline and at follow-up. Three studies used either three^{25,27,28} or two^{26,29} 24-hour recall telephone interviews. The study by de Ruyter et al.²⁴ assessed baseline intake with a questionnaire for the parents and then tracked intake by recollecting the beverage cans they had provided and calculating intake based on whether they were empty, half-empty, or still full. For participants, substituting SSB with non-caloric drinks the mean reduction in kcal/day from beverages during the intervention period was 240.1 kcal/day assessed with the questionnaires specified above (the calculations are specified in Table S4). In all included intervention groups, the participants reported reductions in the total energy intake that were larger than the energy reduction attributed to the

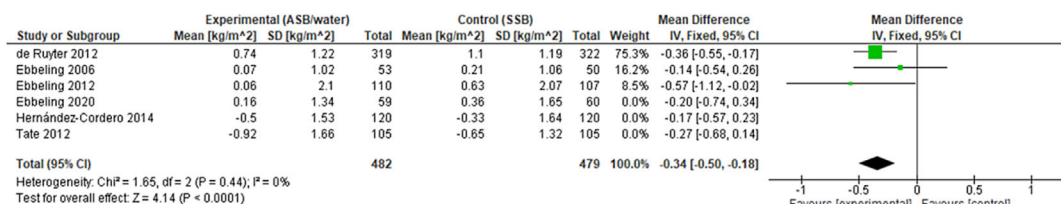


FIGURE 4 Subgroup analysis of children randomized to interventions substituting an existing use of SSB with non-caloric alternatives. Forest plot of change in BMI (kg/m²) in control- (SSB-) and intervention- (ASB and/or water-) groups.

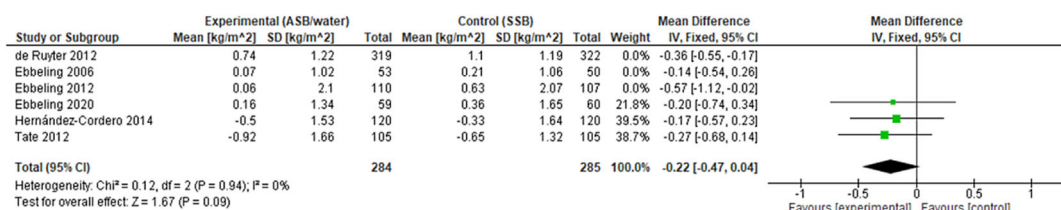


FIGURE 5 Subgroup analysis of adults randomized to control or interventions substituting an existing use of SSB with non-caloric alternatives. Forest plot of change in BMI (kg/m²) in control- (SSB-) and intervention- (ASB and/or water-) groups. In studies with multiple intervention groups, the group that obtained the largest effect is shown.

decreased intake of SSBs. Of notice, most control groups also reported substantial reductions in their total energy intake.

3.8 | Certainty of evidence

Using the GRADE method, the available evidence was rated as having high certainty (Table S5). The risk of bias was evaluated and estimated to be overall low, and the results were generally consistent showing the same direction of effect, with little variation in effect sizes.

4 | DISCUSSION

Findings from the present systematic review and meta-analysis showed that replacing an existing SSB-consumption with ASBs or water reduced BMI about 0.3 units during 6–18 months interventions. ASBs was as effective as water at reducing weight. The BMI reduction translates into a weight loss of approximately 0.5 and 1 kg for normal weight children and adults, respectively.^{38,39} The only study with follow-up of body weight after the intervention had ended showed a regression towards the baseline BMI 1 year after the intervention had ceased.

The interpretation of results from subgroup analyses must be done with caution due to the limited number of studies included in the analyses and the non-significant findings. With this condition in mind, we found a trend towards a larger BMI reduction in interventions of long duration (>12 months). However, the study by de Ruyter et al.,³⁷ which was both the longest and largest of the studies included, found that the bulk of the BMI change took place within the first 6 months. We also found a trend towards a larger BMI reduction in studies including children compared with interventions among adults. Even though this finding was also to some degree explained by the impact of the large study by de Ruyter et al.,²⁴ it may indicate that children's BMI is influenced more by SSBs than the BMI of adults.

There were no major differences in effect size in the subgroup analyses of studies differing in baseline BMI and handling of missing data. Still, it was unexpected that the point estimate of weight loss was larger in studies using multiple imputation than in studies using completers' analysis. The finding indicate that the algorithm used for multiple imputation inflated the results. Still, only three studies were included in each group, the confidence intervals overlap, and the finding is most likely a matter of chance.

The three original articles by Ebbeling et al. all described effect modification related to phenotype, with a greater BMI reduction in Hispanics,²⁵ in the tertile of participants with highest baseline BMI²⁶ and among the participants with the highest amount of trunk fat at baseline.²⁷ de Ruyter et al. also described greater reduction in BMI with higher baseline BMI.⁴⁰ Altogether, the original subgroup analyses from the included trials indicate that the magnitude of the BMI change due to replacing SSBs with non-caloric beverages may vary on factors such as ethnicity,²⁵ baseline BMI^{24,26} and fat

distribution.²⁷ Of other important effect modifiers should be mentioned: socio-economic position,⁴¹ genetics,¹¹ perceived stress and perceived discrimination.⁴²

All included studies assessed adherence and food intake in some way, most studies used 24-hour recall interviews. In all included intervention groups, participants reported reductions in the total energy intake that were larger than the energy reduction attributed to the decreased intake of SSBs. However, bias in reporting energy intake is notorious, which is why randomized studies are warranted in nutritional epidemiology. Still, the reported reduction in energy intake from SSBs (240.1 kcal) was equivalent to approximately 0.5 L of SSBs; a reduction in the same order of magnitude as the interventions aimed for. Thus, the interventions appear to have been efficient in reducing the intake of SSBs. An often-used naive calculation suggests 7.700 kcal per kg weight loss⁴³ equivalent to a 5.6 kg weight loss in 6 months for the (adult) participants in the included intervention groups (for calculation, see [Supporting Information](#)). The mean weight loss achieved across the included studies was considerably less, and the results point to that naive calculations of weight change are as misleading in practice as they are from a theoretical perspective.⁴⁴ The results indicate that the reduced intake of SSBs was either compensated for by intake of other foods or compensated for by reduced energy expenditure.

4.1 | Strengths and weaknesses

We consider it a strength that the present study included only “substitution” design RCTs. Earlier meta-analyses in the field have mixed substitution designs with overfeeding designs making them difficult to interpret and resulting in a high degree of heterogeneity as assessed with I^2 . In all our analyses, I^2 was 0. The substitution design reflects a common situation in the clinic and in people's everyday life: *How much weight can I expect to lose if I replace my SSB-consumption with ASBs or water?* Thus, the results present research results that relatively easy translates into real life and public health advice. Furthermore, it is a substantial strength that all included studies were of relatively long duration and had low risk of bias, including low risk of publication bias.

Still, including only studies with long follow-up and a substitution design was at the same time a limitation as only six studies were eligible, of which half were by the same first author. However, the Risk of Bias-evaluation showed low risk of bias, including the studies by Ebbeling et al.

Especially our subgroup analyses should be interpreted with caution as at least 10 studies should be included in each subgroup, according to the Cochrane Handbook.⁴⁵

4.2 | Comparing results with other literature

Prospective cohorts consistently show an association between SSB-consumption and high BMI.² But, there is also an association between

ASB-consumption and high BMI,¹⁴ likely pointing to the limitations of the cohort studies in regard to residual confounding and reverse causation. Altogether, cohort studies have not convincingly demonstrated an association between SSB intake and weight change. As one of few areas in nutrition research, numerous RCTs have been conducted about soft drink consumption. Due to the well-known bias in nutritional epidemiology, we consider the results from RCTs as more trustworthy in gaining knowledge in this field of nutrition research.

A previous systematic review from 2013 by Malik et al.,⁴ included both prospective cohorts studies and trials. The authors found similar trends and effects in the same order of magnitude as we found in this review: increasing the SSB-consumption led to weight gain and replacing it with non-caloric drinks led to a small weight loss or a smaller weight gain. Even though other meta-analyses have yielded somewhat different results,⁴⁶ the majority conclude that SSBs do contribute to weight gain.⁸ These findings are indirectly in line with our main finding: that substitution of SSBs with non-caloric drinks led to a small weight loss or a smaller weight gain as long as the intervention continued.

4.3 | Perspectives for public health action

Taxation of SSB have often been suggested as a way for structural prevention of weight gain. Taxation may work well to reduce sales of SSB, but our analyses point to that a reduced consumption of SSBs does not proportionally translate into weight reduction. The explanation for this is likely that a reduced intake of SSBs is compensated for by reduced metabolic energy expenditure or increased intake of calories from other sources. A weight loss of 0.3 BMI points may not be clinically relevant on an individual level, but it is a rather large effect compared with other more complex weight loss interventions.³⁰ High BMI is a multifactorial condition, and SSB consumption is just one of many contributing factors to weight gain. Even though replacing SSB consumption with non-caloric alternatives will likely have a very modest effect on body weight on the individual level, targeting SSBs through policies such as taxation, reducing visibility of SSBs, and reducing access to SSBs in schools might be meaningful public health interventions on the population-level.^{8,47}

In terms of health, drinking water is probably superior to other drinks, but ASBs might play a role in the transition from SSBs to water.⁸ It is complicated to make general recommendations for ASB consumption as different artificial sweeteners differ substantially in chemical structure and effect.^{19,48} Despite the diverse nature of the artificial sweeteners, they are often pooled in one single group in meta-analyses, and more studies of the individual substances and consequences of long-term use is needed.⁴⁹ Still, for now, they appear better than the caloric alternative in regard to preventing weight gain or obtaining a small weight loss.⁵⁰ Future research is warranted on how to increase global access to potable water and long-term health effects of consuming ASBs.

5 | CONCLUSION

Replacing SSB-consumption with non-caloric drinks resulted in a small but statistically significant long-term (as long as the interventions lasted) reduction in body weight of approximately 0.3 BMI units equivalent to approximately 0.5 and 1 kg in normal weight in children and adults, respectively.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest. There was no financial support for the study.

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REFERENCES

1. Bucher Della Torre S, Keller A, Laure Depeyre J, Kruseman M. Sugar-sweetened beverages and obesity risk in children and adolescents: a systematic analysis on how methodological quality may influence conclusions. *J Acad Nutr Diet*. 2016;116(4):638-659. doi:[10.1016/j.jand.2015.05.020](https://doi.org/10.1016/j.jand.2015.05.020)
2. Hu FB. Resolved: there is sufficient scientific evidence that decreasing sugar-sweetened beverage consumption will reduce the prevalence of obesity and obesity-related diseases. *Obes Rev*. 2013;14(8):606-619. doi:[10.1111/obr.12040](https://doi.org/10.1111/obr.12040)
3. Kaiser KA, Shikany JM, Keating KD, Allison DB. Will reducing sugar-sweetened beverage consumption reduce obesity? Evidence supporting conjecture is strong, but evidence when testing effect is weak. *Obes Rev*. 2013;14(8):620-633. doi:[10.1111/obr.12048](https://doi.org/10.1111/obr.12048)
4. Malik VS, Pan A, Willett WC, Hu FB. Sugar-sweetened beverages and weight gain in children and adults: a systematic review and meta-analysis. *Am J Clin Nutr*. 2013;98(4):1084-1102. doi:[10.3945/ajcn.113.058362](https://doi.org/10.3945/ajcn.113.058362)
5. Mattes RD, Shikany JM, Kaiser KA, Allison DB. Nutritively sweetened beverage consumption and body weight: a systematic review and meta-analysis of randomized experiments. *Obes Rev*. 2011;12(5):346-365. doi:[10.1111/j.1467-789X.2010.00755.x](https://doi.org/10.1111/j.1467-789X.2010.00755.x)
6. DiMeglio DP, Mattes RD. Liquid versus solid carbohydrate: effects on food intake and body weight. *Int J Obes Relat Metab Disord*. 2000;24(6):794-800. doi:[10.1038/sj.jjo.0801229](https://doi.org/10.1038/sj.jjo.0801229)
7. Guthrie JF, Morton JF. Food sources of added sweeteners in the diets of Americans. *J Am Diet Assoc*. 2000;100(1):43-51. doi:[10.1016/S0002-8223\(00\)00018-3](https://doi.org/10.1016/S0002-8223(00)00018-3)
8. Malik VS, Hu FB. The role of sugar-sweetened beverages in the global epidemics of obesity and chronic diseases. *Nat Rev Endocrinol*. 2022;18(4):205-218. doi:[10.1038/s41574-021-00627-6](https://doi.org/10.1038/s41574-021-00627-6)
9. Pan A, Hu FB. Effects of carbohydrates on satiety: differences between liquid and solid food. *Curr Opin Clin Nutr Metab Care*. 2011;14(4):385-390. doi:[10.1097/MCO.0b013e328346df36](https://doi.org/10.1097/MCO.0b013e328346df36)
10. Reedy J, Krebs-Smith SM. Dietary sources of energy, solid fats, and added sugars among children and adolescents in the United States. *J Am Diet Assoc*. 2010;110(10):1477-1484. doi:[10.1016/j.jada.2010.07.010](https://doi.org/10.1016/j.jada.2010.07.010)
11. Qi Q, Chu AY, Kang JH, et al. Sugar-sweetened beverages and genetic risk of obesity. *N Engl J Med*. 2012;367(15):1387-1396. doi:[10.1056/NEJMoa1203039](https://doi.org/10.1056/NEJMoa1203039)

12. Nordmo M, Danielsen YS, Nordmo M. The challenge of keeping it off, a descriptive systematic review of high-quality, follow-up studies of obesity treatments. *Obes Rev*. 2020;21(1):e12949. doi:[10.1111/obr.12949](https://doi.org/10.1111/obr.12949)
13. Trumbo PR, Rivers CR. Systematic review of the evidence for an association between sugar-sweetened beverage consumption and risk of obesity. *Nutr Rev*. 2014;72(9):566-574. doi:[10.1111/nure.12128](https://doi.org/10.1111/nure.12128)
14. Ruanpeng D, Thongprayoon C, Cheungpasitporn W, Harindhanavudhi T. Sugar and artificially sweetened beverages linked to obesity: a systematic review and meta-analysis. *Qjm*. 2017;110(8):513-520. doi:[10.1093/qjmed/hcx068](https://doi.org/10.1093/qjmed/hcx068)
15. Forshee RA, Anderson PA, Storey ML. Sugar-sweetened beverages and body mass index in children and adolescents: a meta-analysis. *Am J Clin Nutr*. 2008;87(6):1662-1671. doi:[10.1093/ajcn/87.6.1662](https://doi.org/10.1093/ajcn/87.6.1662)
16. Te Morenga L, Mallard S, Mann J. Dietary sugars and body weight: systematic review and meta-analyses of randomised controlled trials and cohort studies. *BMJ*. 2012;346(jan15 3):e7492. doi:[10.1136/bmj.e7492](https://doi.org/10.1136/bmj.e7492)
17. Vartanian LR, Schwartz MB, Brownell KD. Effects of soft drink consumption on nutrition and health: a systematic review and meta-analysis. *Am J Public Health*. 2007;97(4):667-675. doi:[10.2105/AJPH.2005.083782](https://doi.org/10.2105/AJPH.2005.083782)
18. Tordoff MG, Alleva AM. Effect of drinking soda sweetened with aspartame or high-fructose corn syrup on food intake and body weight. *Am J Clin Nutr*. 1990;51(6):963-969. doi:[10.1093/ajcn/51.6.963](https://doi.org/10.1093/ajcn/51.6.963)
19. Higgins KA, Mattes RD. A randomized controlled trial contrasting the effects of 4 low-calorie sweeteners and sucrose on body weight in adults with overweight or obesity. *Am J Clin Nutr*. 2019;109(5):1288-1301. doi:[10.1093/ajcn/nqy381](https://doi.org/10.1093/ajcn/nqy381)
20. Njike VY, Faridi Z, Shuval K, et al. Effects of sugar-sweetened and sugar-free cocoa on endothelial function in overweight adults. *Int J Cardiol*. 2011;149(1):83-88. doi:[10.1016/j.ijcard.2009.12.010](https://doi.org/10.1016/j.ijcard.2009.12.010)
21. Reid M, Hammersley R, Duffy M. Effects of sucrose drinks on macronutrient intake, body weight, and mood state in overweight women over 4 weeks. *Appetite*. 2010;55(1):130-136. doi:[10.1016/j.appet.2010.05.001](https://doi.org/10.1016/j.appet.2010.05.001)
22. Maersk M, Belza A, Stødkilde-Jørgensen H, et al. Sucrose-sweetened beverages increase fat storage in the liver, muscle, and visceral fat depot: a 6-mo randomized intervention study. *Am J Clin Nutr*. 2012;95(2):283-289. doi:[10.3945/ajcn.111.022533](https://doi.org/10.3945/ajcn.111.022533)
23. Raben A, Vasilaras TH, Møller AC, Astrup A. Sucrose compared with artificial sweeteners: different effects on ad libitum food intake and body weight after 10 wk of supplementation in overweight subjects. *Am J Clin Nutr*. 2002;76(4):721-729. doi:[10.1093/ajcn/76.4.721](https://doi.org/10.1093/ajcn/76.4.721)
24. de Ruyter JC, Olthof MR, Seidell JC, Katan MB. A trial of sugar-free or sugar-sweetened beverages and body weight in children. *N Engl J Med*. 2012;367(15):1397-1406. doi:[10.1056/NEJMoa1203034](https://doi.org/10.1056/NEJMoa1203034)
25. Ebbeling CB, Feldman HA, Chomitz VR, et al. A randomized trial of sugar-sweetened beverages and adolescent body weight. *N Engl J Med*. 2012;367(15):1407-1416. doi:[10.1056/NEJMoa1203388](https://doi.org/10.1056/NEJMoa1203388)
26. Ebbeling CB, Feldman HA, Osganian SK, Chomitz VR, Ellenbogen SJ, Ludwig DS. Effects of decreasing sugar-sweetened beverage consumption on body weight in adolescents: a randomized, controlled pilot study. *Pediatrics*. 2006;117(3):673-680. doi:[10.1542/peds.2005-0983](https://doi.org/10.1542/peds.2005-0983)
27. Ebbeling CB, Feldman HA, Steltz SK, Quinn NL, Robinson LM, Ludwig DS. Effects of sugar-sweetened, artificially sweetened, and unsweetened beverages on Cardiometabolic risk factors, body composition, and sweet taste preference: a randomized controlled trial. *J Am Heart Assoc*. 2020;9(15):e015668. doi:[10.1161/JAHA.119.015668](https://doi.org/10.1161/JAHA.119.015668)
28. Hernández-Cordero S, Barquera S, Rodríguez-Ramírez S, et al. Substituting water for sugar-sweetened beverages reduces circulating triglycerides and the prevalence of metabolic syndrome in obese but not in overweight Mexican women in a randomized controlled trial. *J Nutr*. 2014;144(11):1742-1752. doi:[10.3945/jn.114.193490](https://doi.org/10.3945/jn.114.193490)
29. Tate DF, Turner-McGrievy G, Lyons E, et al. Replacing caloric beverages with water or diet beverages for weight loss in adults: main results of the choose healthy options consciously everyday (CHOICE) randomized clinical trial. *Am J Clin Nutr*. 2012;95(3):555-563. doi:[10.3945/ajcn.111.026278](https://doi.org/10.3945/ajcn.111.026278)
30. Madigan CD, Graham HE, Sturgiss E, et al. Effectiveness of weight management interventions for adults delivered in primary care: systematic review and meta-analysis of randomised controlled trials. *BMJ*. 2022;377:e069719. doi:[10.1136/bmj-2021-069719](https://doi.org/10.1136/bmj-2021-069719)
31. Covidence systematic review software, Veritas Health Innovation, Melbourne, Australia. <https://www.covidence.org>
32. The Endnote Team. *Endnote*. Clarivate; 2013.
33. Review Manager (RevMan) [Computer program]. Version 5.4, The Cochrane Collaboration, 2020.
34. Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ*. 2003;327(7414):557-560. doi:[10.1136/bmj.327.7414.557](https://doi.org/10.1136/bmj.327.7414.557)
35. Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ*. 2019;366:l4898. doi:[10.1136/bmj.l4898](https://doi.org/10.1136/bmj.l4898)
36. Sterne JA, Sutton AJ, Ioannidis JP, et al. Recommendations for examining and interpreting funnel plot asymmetry in meta-analyses of randomised controlled trials. *BMJ*. 2011;343(jul22 1):d4002. doi:[10.1136/bmj.d4002](https://doi.org/10.1136/bmj.d4002)
37. de Ruyter JC, Olthof MR, Kuijper LD, Katan MB. Effect of sugar-sweetened beverages on body weight in children: design and baseline characteristics of the double-blind, randomized INtervention study in kids. *Contemp Clin Trials*. 2012;33(1):247-257. doi:[10.1016/j.cct.2011.10.007](https://doi.org/10.1016/j.cct.2011.10.007)
38. Tinggaard JAL, Sorensen K, Peterson JH, Skakkebaek NE, Juul A. Vækstkurve for piger 0 til 20 år. <https://www.sundhed.dk/borger/patienthaandbogen/boern/illustrationer/tegning/vaekstkurver-piger-0-20/>
39. Tinggaard JSK, Aksglaede L, Peterson JH, Juul A. Vækstkurve for drenge 0 til 20 år. <https://www.sundhed.dk/borger/patienthaandbogen/boern/illustrationer/tegning/vaekstkurve-drenge-0-20/>
40. Katan MB, de Ruyter JC, Kuijper LD, Chow CC, Hall KD, Olthof MR. Impact of masked replacement of sugar-sweetened with sugar-free beverages on body weight increases with initial BMI: secondary analysis of data from an 18 month double-blind trial in children. *PLoS ONE*. 2016;11(7):e0159771. doi:[10.1371/journal.pone.0159771](https://doi.org/10.1371/journal.pone.0159771)
41. Stamatakis E, Primatesta P, Chinn S, Rona R, Falaschetti E. Overweight and obesity trends from 1974 to 2003 in English children: what is the role of socioeconomic factors? *Arch Dis Child*. 2005;90(10):999-1004. doi:[10.1136/adc.2004.068932](https://doi.org/10.1136/adc.2004.068932)
42. Sutin AR, Terracciano A. Perceived weight discrimination and obesity. *PLoS ONE*. 2013;8(7):e70048. doi:[10.1371/journal.pone.0070048](https://doi.org/10.1371/journal.pone.0070048)
43. Bray GA, ed. Pathogenesis of obesity: food intake. *Major Problems in Internal Medicine*. 1976;9:44-5.
44. Thomas DM, Ciesla A, Levine JA, Stevens JG, Martin CK. A mathematical model of weight change with adaptation. *Math Biosci Eng*. 2009;6(4):873-887. doi:[10.3934/mbe.2009.6.873](https://doi.org/10.3934/mbe.2009.6.873)
45. Higgins JPT, Thomas J, Chandler J, et al. *Cochrane handbook for systematic reviews of interventions version 6.3 (updated February 2022)*. Cochrane; 2022. <https://www.training.cochrane.org/handbook>
46. Nissensohn M, Fuentes Lugo D, Serra-Majem L. Sugar-sweetened beverage consumption and obesity in children's meta-analyses: reaching wrong answers for right questions. *Nutr Hosp*. 2018;35(2):474-488. doi:[10.20960/nh.1492](https://doi.org/10.20960/nh.1492)

47. Ebbeling CB, Pawlak DB, Ludwig DS. Childhood obesity: public-health crisis, common sense cure. *Lancet*. 2002;360(9331):473-482. doi:[10.1016/S0140-6736\(02\)09678-2](https://doi.org/10.1016/S0140-6736(02)09678-2)
48. Hunter SR, Reister EJ, Cheon E, Mattes RD. Low calorie sweeteners differ in their physiological effects in humans. *Nutrients*. 2019;11(11):2717. doi:[10.3390/nu11112717](https://doi.org/10.3390/nu11112717)
49. Johnson RK, Lichtenstein AH, Anderson CAM, et al. Low-calorie sweetened beverages and Cardiometabolic health: a science advisory from the American Heart Association. *Circulation*. 2018;138(9):e126-e140. doi:[10.1161/CIR.0000000000000569](https://doi.org/10.1161/CIR.0000000000000569)
50. Nadolsky KZ. COUNTERPOINT: artificial sweeteners for obesity-better than sugary alternatives, potentially a solution. *Endocr Pract*. 2021;27(10):1056-1061. doi:[10.1016/j.eprac.2021.06.013](https://doi.org/10.1016/j.eprac.2021.06.013)

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